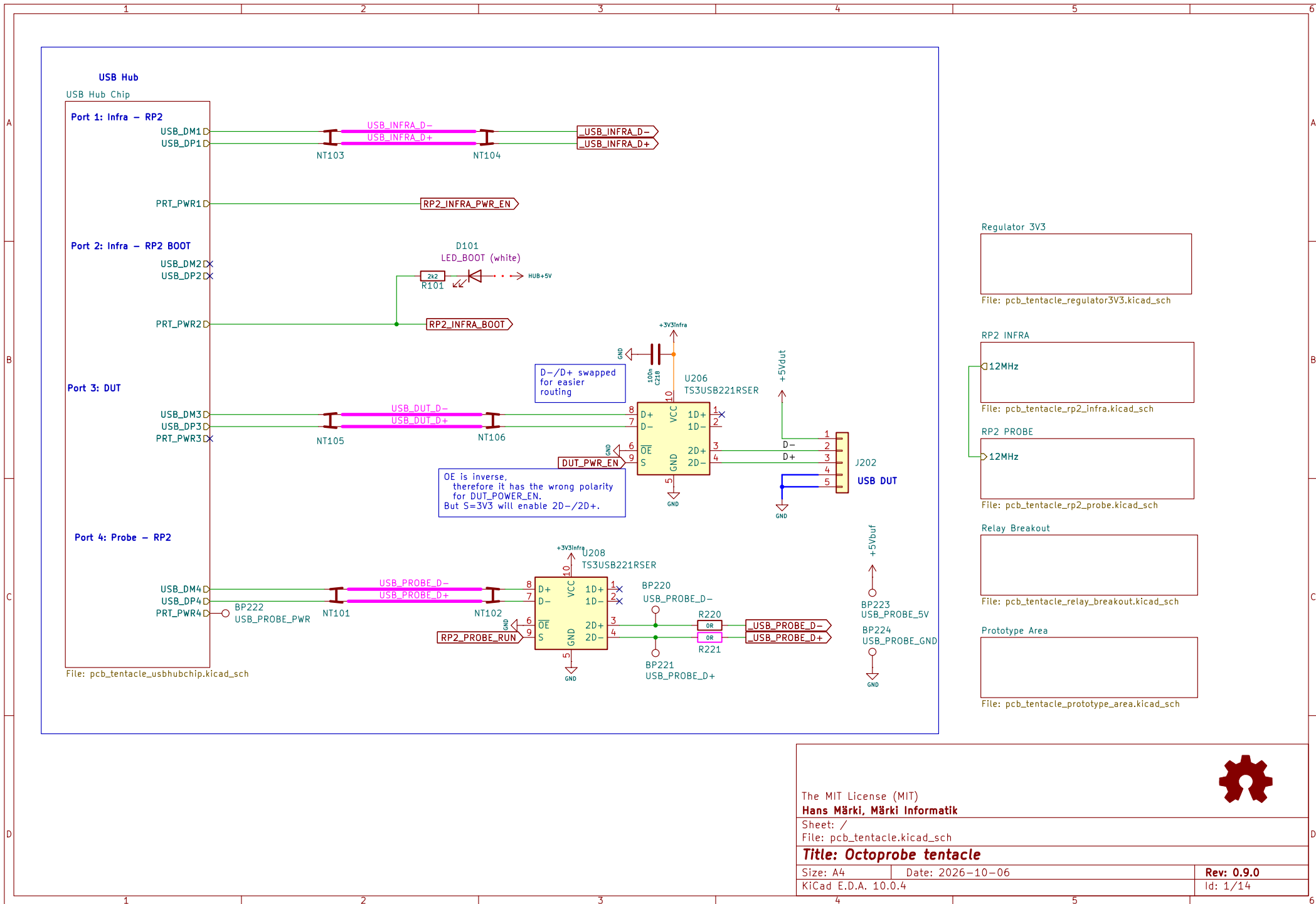
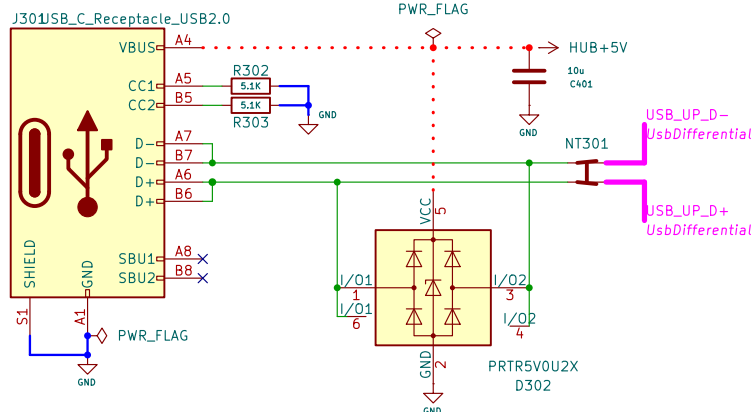


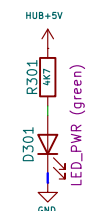


USB C Female

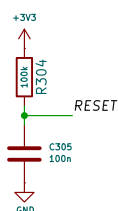
USB Type-C DATA



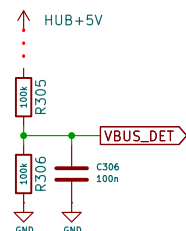
Power LED



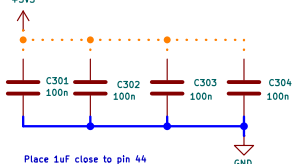
Boot Sequence



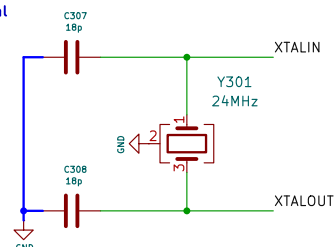
VBUS_DET



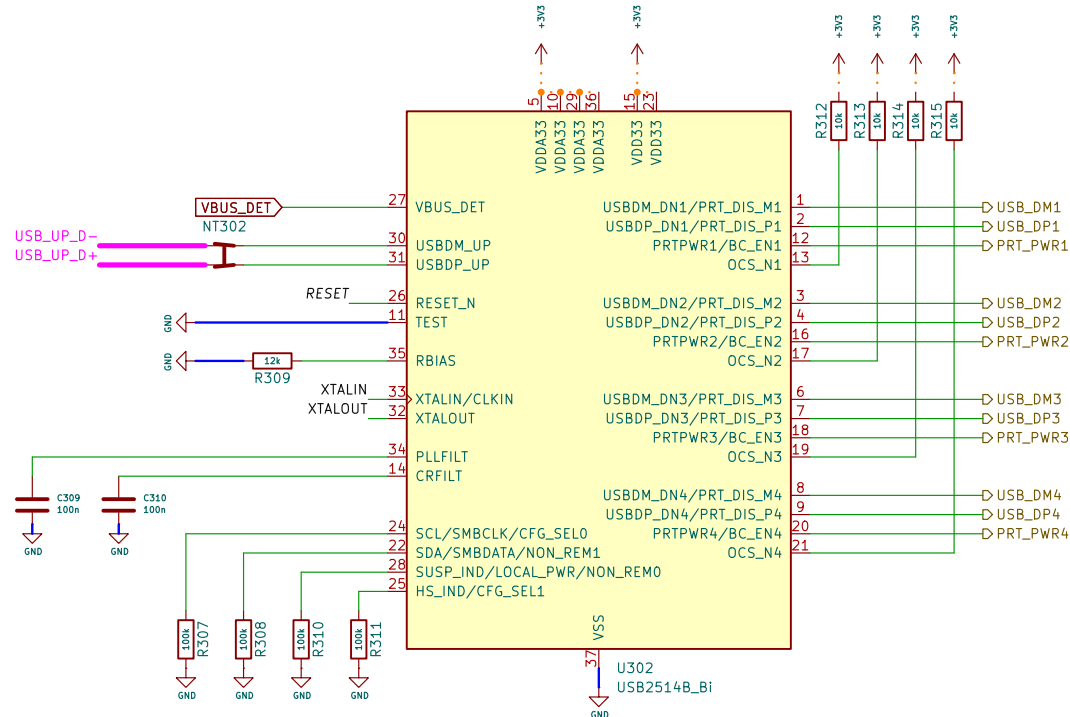
Bypass Capacitors



Crystal



Hub



Scope GND Pad



Microchip AN 15.17 "PCB Layout Guide for USB Hubs"

DP/DM:
27mil, 0.7mm, trace
5mil, 0.127mm, space
50mil, 1.27mm, trace side no ground

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Sheet: /USB Hub Chip/

File: pcb_tentacle_usbhubchip.kicad_sch

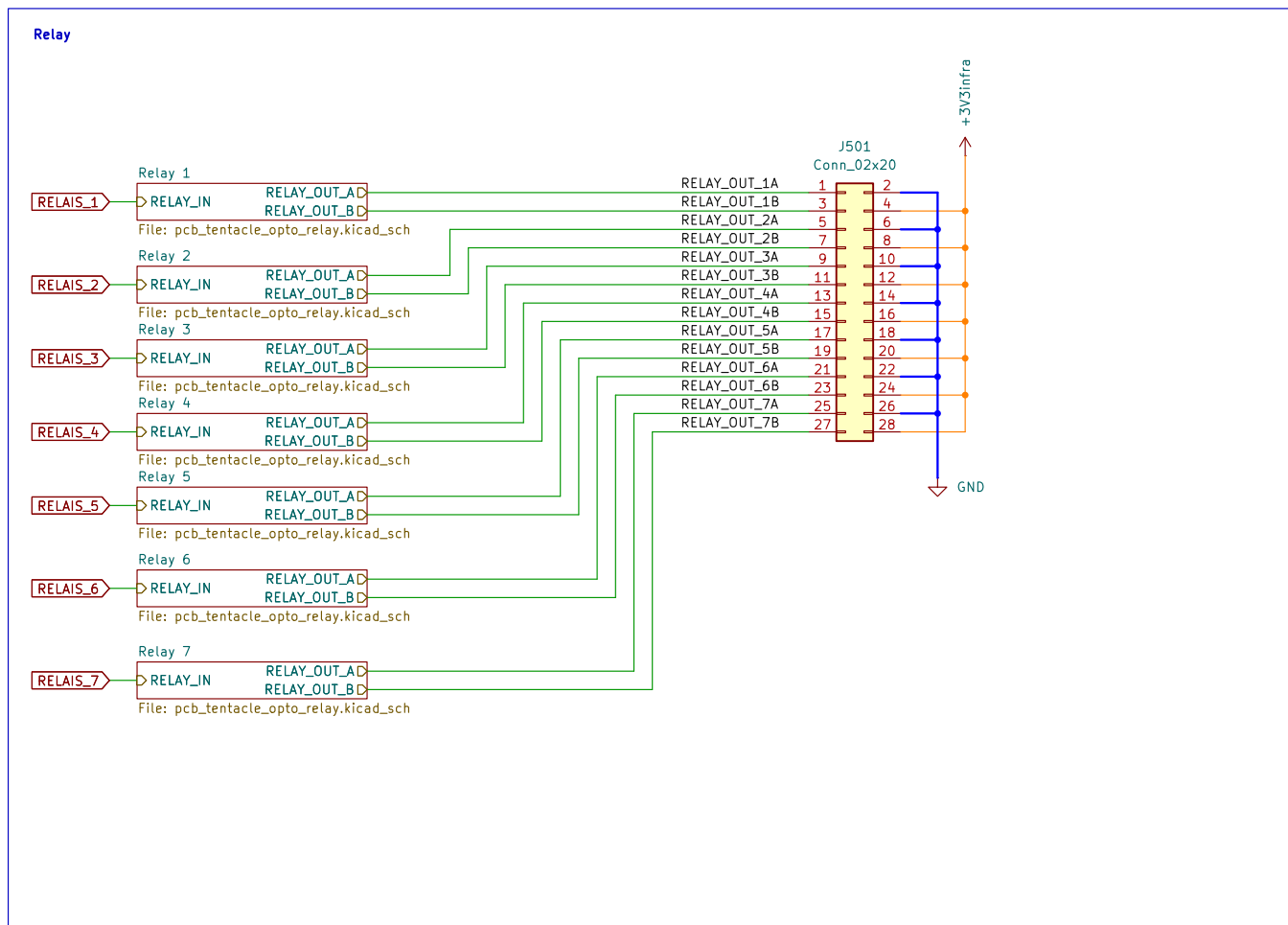
Title: Octoprobe tentacle

Size: A4 Date: 2026-10-06

KiCad E.D.A. 10.0.4

Rev: 0.9.0

Id: 3/14



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Hans Märki, Märki Informatik

Sheet: /Relay Breakout/

File: pcb_tentacle_relay_breakout.kicad_sch

Title: Octoprobe tentacle

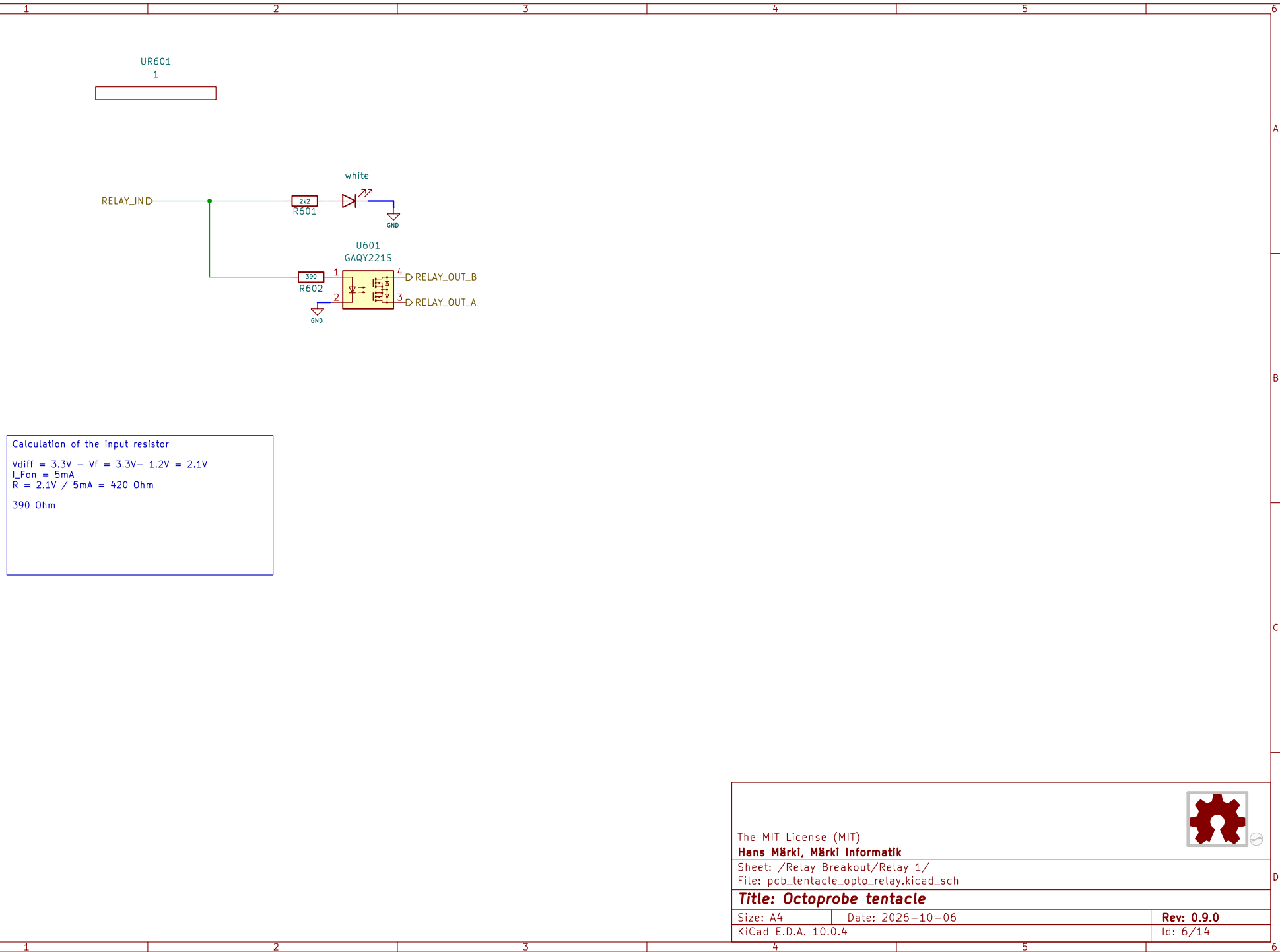
Size: A4

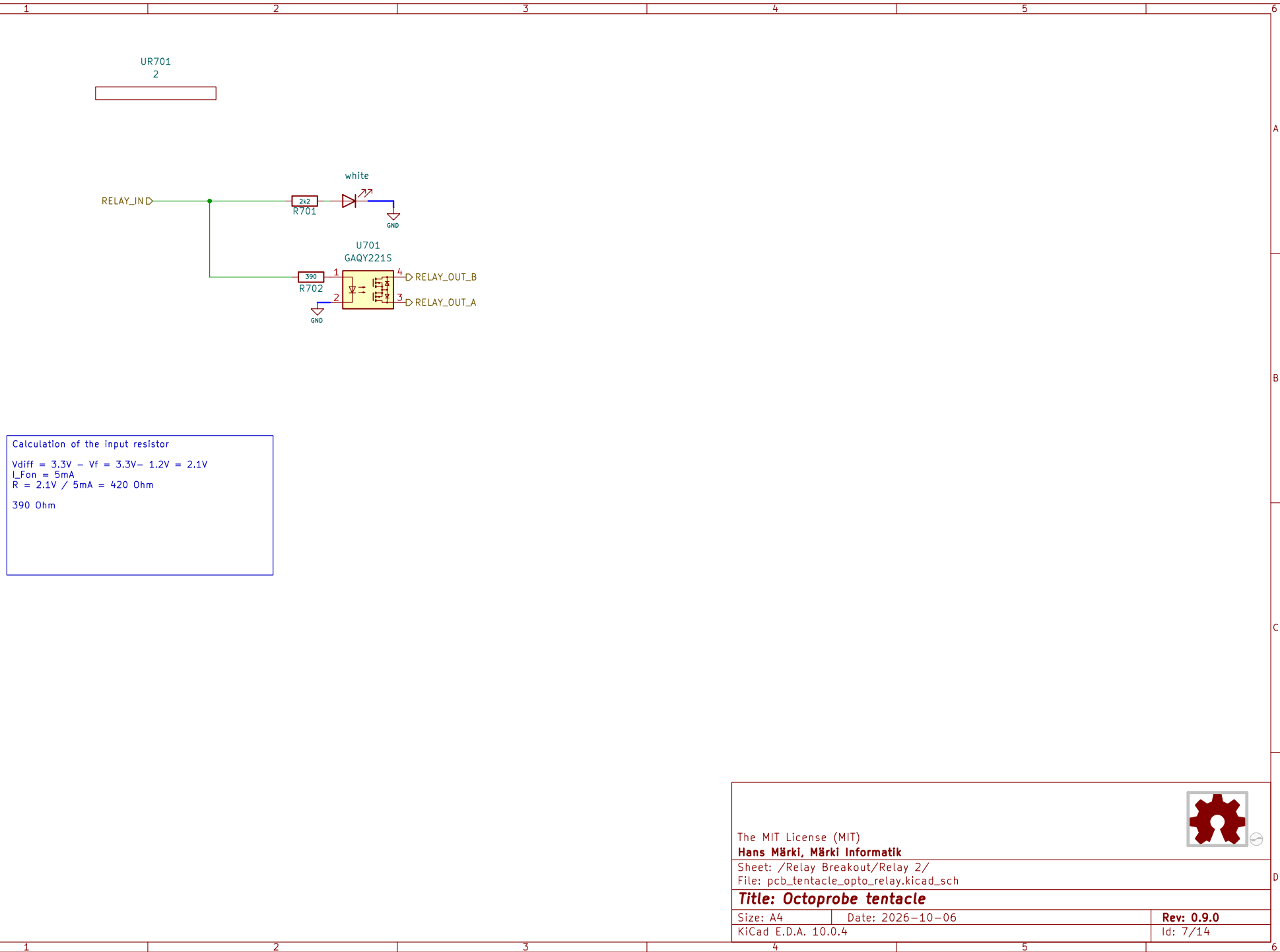
Date: 2026-10-06

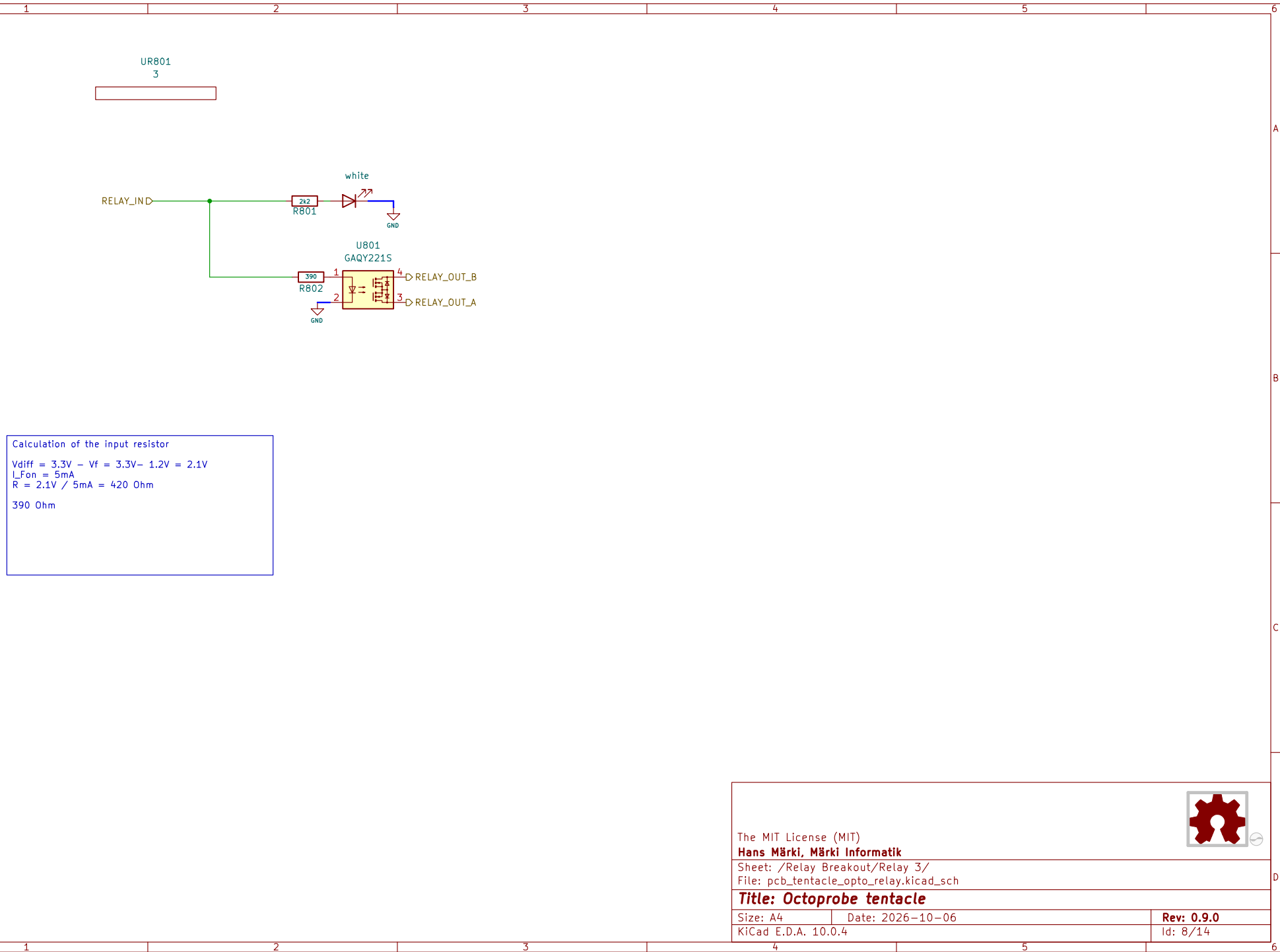
Rev: 0.9.0

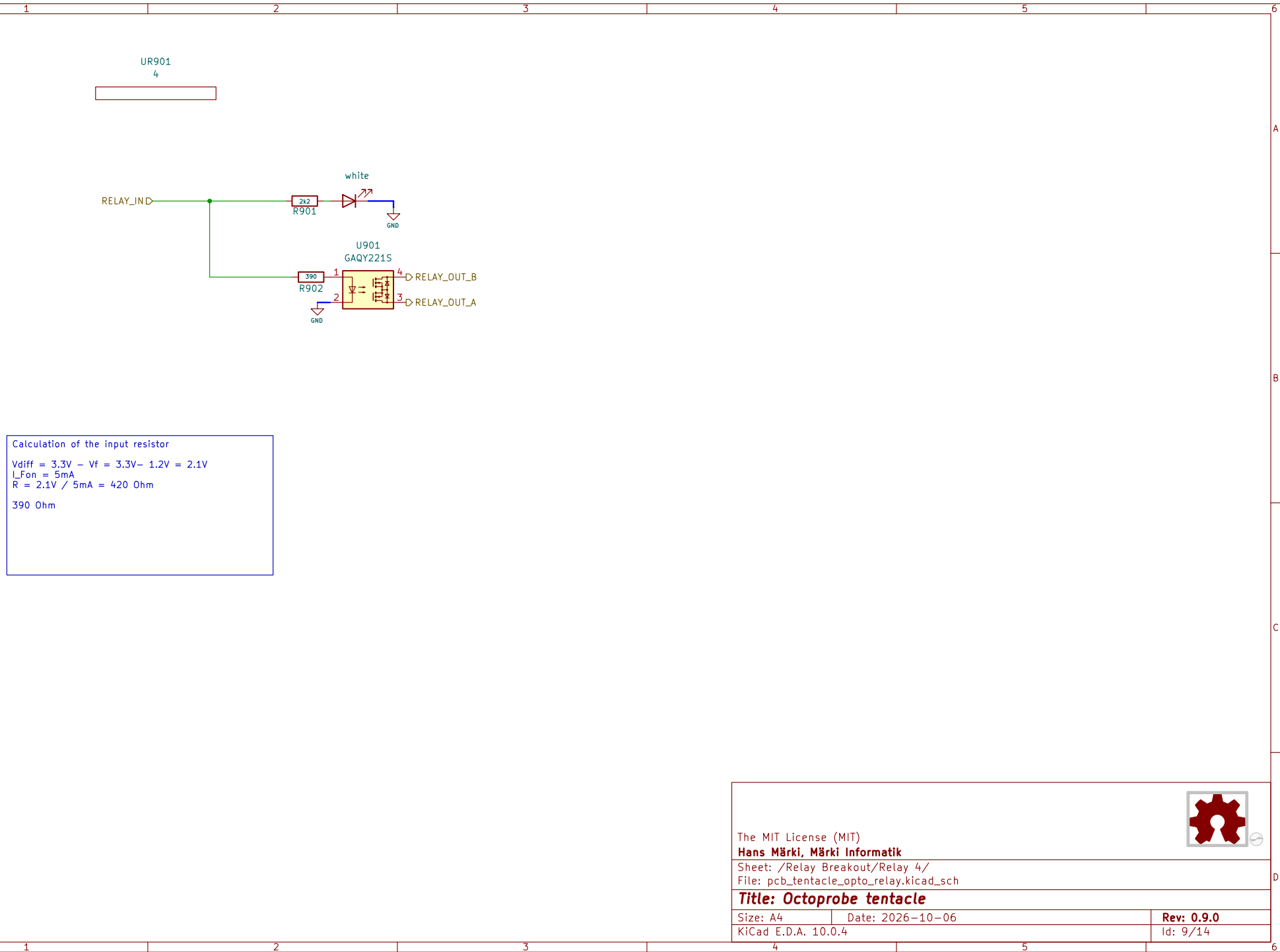
KiCad E.D.A. 10.0.4

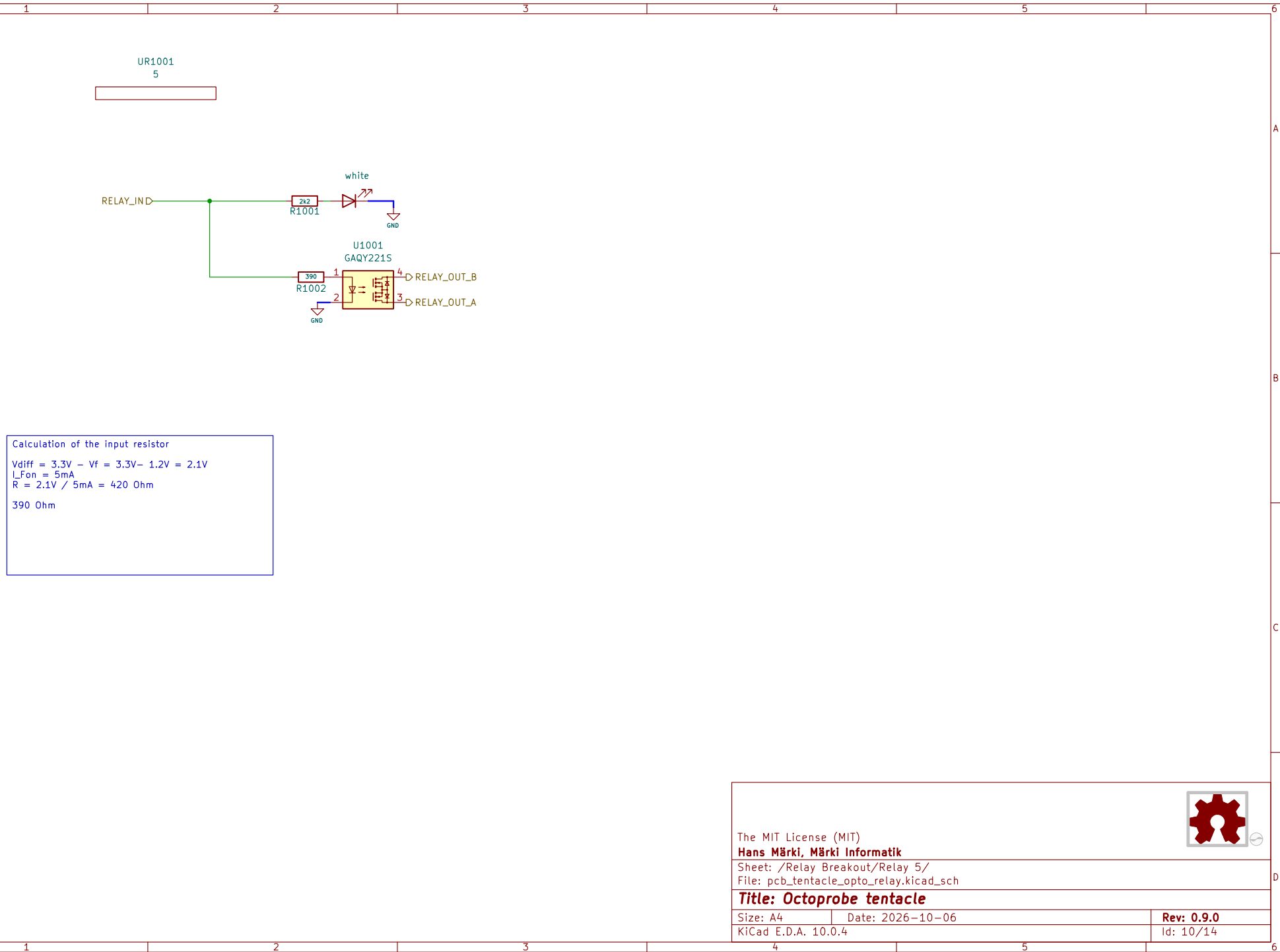
Id: 5/14

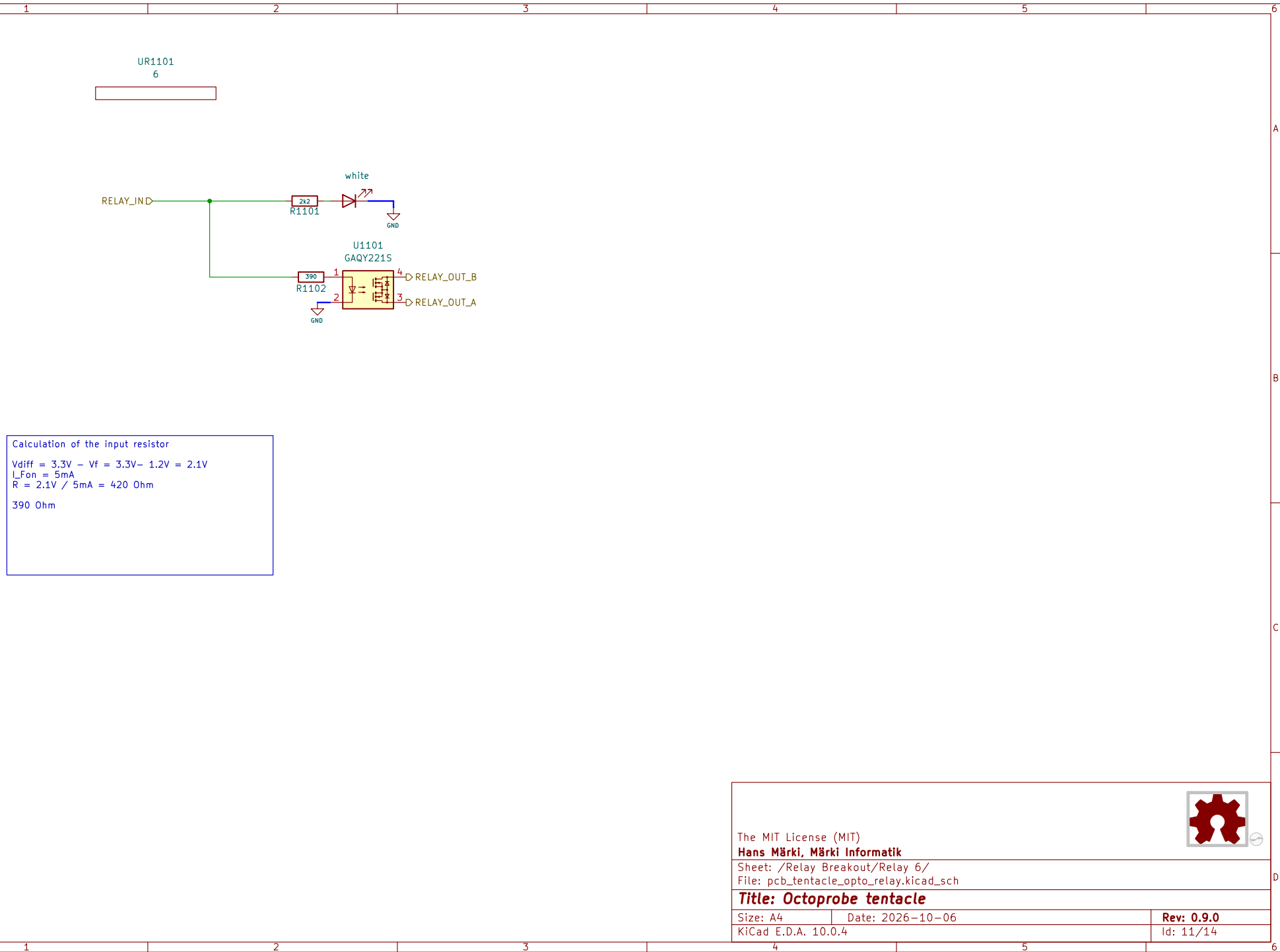


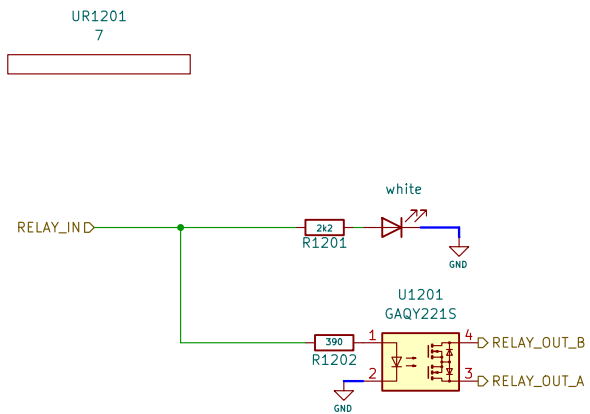












Calculation of the input resistor

$V_{diff} = 3.3V - V_f = 3.3V - 1.2V = 2.1V$

$I_{Fon} = 5mA$

$R = 2.1V / 5mA = 420 \text{ Ohm}$

390 Ohm



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Sheet: /Relay Breakout/Relay 7/

File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4

Date: 2026-10-06

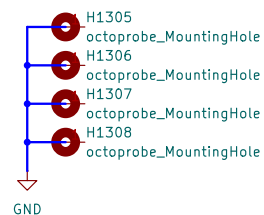
Rev: 0.9.0

KiCad E.D.A. 10.0.4

Id: 12/14

Prototype Area

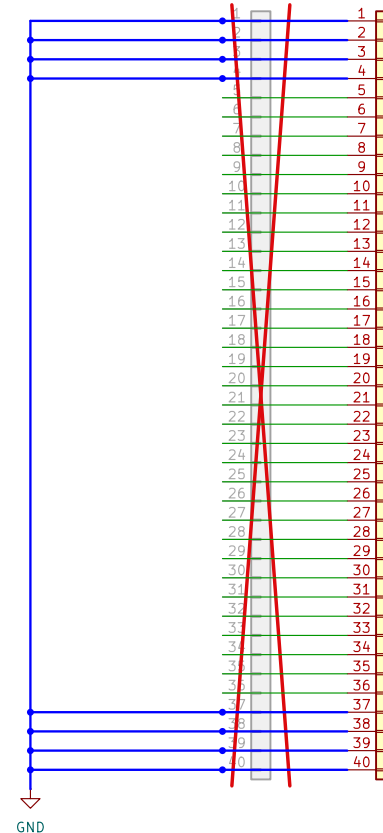
- TH1301
JLPCB Tooling Hole
- TH1302
JLPCB Tooling Hole



Octobus

J1302
Conn_02x20

J1303
Conn_02x20



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Sheet: /Prototype Area/

File: pcb_tentacle_prototype_area.kicad_sch

Title: Octoprobe tentacle

Size: A4

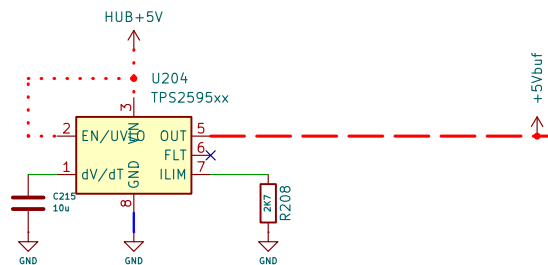
Date: 2026-10-06

Rev: 0.9.0

KiCad E.D.A. 10.0.4

Id: 13/14

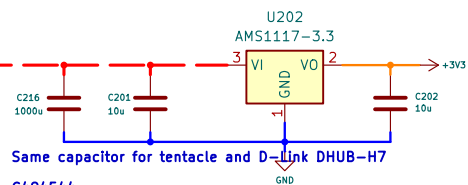
Current Limiter 5V



Datasheet "Active Current Limiting":
Goal: 800mA
 $R_{lim} = 2000 / (0.8A - 0.04) = R2K630$, choosen: R2K7

Datasheet "Slew Rate and Inrush Current Control":
Goal: 5V/1000ms
 $C = 42000 / V \cdot ms = 6'363'636pF = 6uF$, choosen: 10uF

Regulator 3V3

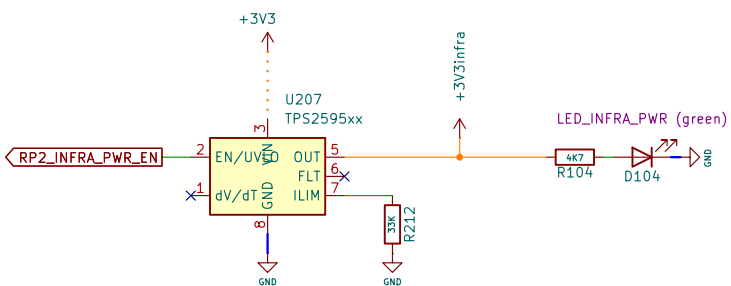


Same capacitor for tentacle and D-Link DHUB-H7

C494544

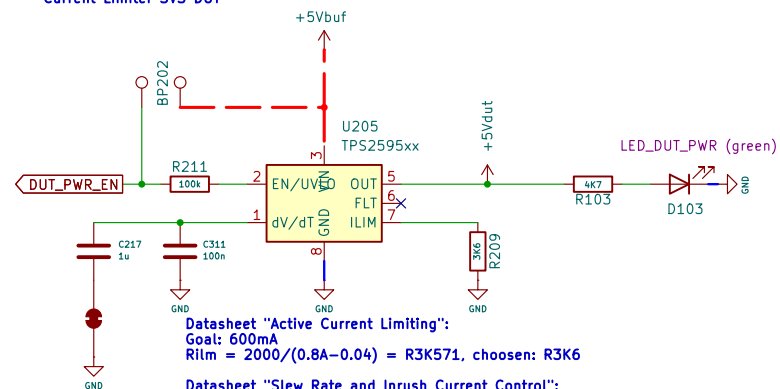
2.5mm distance between wires
6.3mm diameter
11mm height

POWER RP Infra



Datasheet "Active Current Limiting":
Goal: 100mA
 $R_{lim} = 2000 / (0.1A - 0.04) = R33K333$, choosen: R33K

Current Limiter 3V3 DUT



Datasheet "Active Current Limiting":
Goal: 600mA
 $R_{lim} = 2000 / (0.8A - 0.04) = R3K571$, choosen: R3K6

Datasheet "Slew Rate and Inrush Current Control":
Goal: 5V/150ms: 1uF
 $C = 42'000 / V \cdot ms = 1'272'272pF$, choosen: 1uF
Goal: 5/15ms: 100nF

Sheet: /Regulator 3V3/
File: pcb_tentacle_regulator3V3.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 10.0.4

Id: 16/14